

JOY OLUSANYA

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EDUCATION

Obafemi Awolowo University, Osun State, Nigeria

January 2021 – September 2025

Bachelor of Arts in Linguistics

Thesis Title: *Tone in Yoruba Automatic Speech Recognition: Evaluating the Impact of Tone Recognition on Transformer-Based ASR Models*

Manuscript accepted to LoResLM Workshop @ EACL 2026

Project Supervisor: Dr. Kolawole Adeniyi

LINGUISTIC SKILLS

- Linguistics & Computational Linguistics (Phonetics, Morphology, Syntax, Semantics)
- Language Documentation, Phonetics, Tone Analysis, Data Validation, Corpus Analysis, Data Annotation, Dialectology
- Research: Literature Review, Experimental Design, Data Collection, Dataset Curation, Statistical Analysis
- Linguistics Software (Praat, ELAN, FLEx, Audacity)
- Language Proficiency: Yoruba, English, Nigerian Pidgin English, German

TOOLS & FRAMEWORKS

• Python (pandas, NumPy, scikit-learn, matplotlib) • Julia • SQL • Deep Learning Frameworks (PyTorch, TensorFlow, Keras, JAX, Scikit-learn) • NLP Language Models (Transformers, Hugging Face, NLTK, spaCy, Machine Translation, ASR, LLM Evaluation) • Model Evaluation (BLEU, ROUGE, BERTScore, COMET, Human Annotation, Benchmarking) • MLOps Tools (Docker, MLflow, W&B, GitHub, AWS, GCP) • Web Frameworks / APIs (Streamlit, FastAPI, Flask, LangChain, OpenAI API, Gemini API, Tavily API)

PUBLICATIONS

Activation Editing for Low-Resource Cross-Lingual Machine Translation: Does JoLA Transfer? ([link](#))

Accepted to the Women in Machine Learning (WiML) Symposium @ ICML 2026, International Conference on Machine Learning, 2026

Tone in Yoruba Automatic Speech Recognition: Evaluating the Impact of Tone Recognition on Transformer-Based ASR Models ([link](#))

Accepted to LoResLM Workshop @ EACL 2026, European Chapter of the Association for Computational Linguistics, 2026

ÒWE-YOR: Leveraging Transformer-Based Models for Yoruba Proverb Classification ([link](#))

Accepted to WiML @ NeurIPS 2025, Women in Machine Learning Workshop, 2025

WORK EXPERIENCE

Tonative

NLP Researcher & Yoruba Language Lead

Lagos, Nigeria

August 2025 – Present

- Developed multilingual NLP datasets for low-resource African languages, ensuring linguistic diversity and cultural relevance.
- Contributed to machine translation research by conducting a machine translation of English into Kikuyu, Kinyarwanda, and Yoruba for multilingual NLP systems.
- Directed language data collection, annotation, validation, and benchmarking workflows to improve model quality and performance.
- Collaborated on multilingual NLP system design to enhance accessibility, inclusivity, and translation quality across African languages.
- Validated English-to-Yorùbá machine translation datasets to improve linguistic accuracy, translation quality, and cultural representation.
- Certification of Contribution: [View Certificate](#)

FriendnPal

AI Developer

Lagos, Nigeria

January 2023 – March 2025

- Developed NLP-powered mental health chatbot systems to improve conversational relevance, contextual understanding, and user engagement.
- Implemented Retrieval-Augmented Generation (RAG) pipelines, preprocessing workflows, and multilingual data systems for scalable AI applications.

- Curated and analysed multilingual datasets to improve model reliability, response quality, and evaluation performance.

Atom Group (Contract)

AI/LLM Developer

Lagos, Nigeria

April 2025

- Developed a commercial AI agent with Instagram scraping pipelines for social media intelligence and automation.
- Built modular API integration and automated data collection systems to improve efficiency and scalability.
- Improved scraping accuracy and reduced system failures through reusable utilities and exception handling.

Brilla AI Project, Kwame AI Ghana

AI Developer

Remote

January 2024 – March 2025

- Developed multilingual speech-to-text and ASR models with accent robustness using deep learning techniques.
- Validated multilingual datasets to improve transcription quality and model reliability.
- Built reproducible preprocessing and evaluation workflows for scalable multilingual speech systems.

Rendo AI Nigeria

ML/NLP Engineer

Lagos, Nigeria

June 2024 – December 2024

- Designed Python-based web scraping pipelines using Requests, Selenium, and BeautifulSoup.
- Automated large-scale data acquisition workflows, improving dataset completeness and operational efficiency.
- Developed reusable scraping and preprocessing tools for structured multilingual datasets.

Cohere AI

Open Source Contributor

Remote

July 2024 – August 2024

- Validated 500+ multilingual translations for AI benchmarks, ensuring linguistic accuracy, consistency, and quality assurance.
- Collaborated with international open-source contributors on multilingual NLP evaluation and benchmarking tasks.
- Certification of Contribution: [View Certificate](#)

LEADERSHIP EXPERIENCE

Center for Low-Resource Languages and Cultures (CLRLC)

Founder & CEO

Nigeria

2025 – Present

- Established a research and advocacy center dedicated to NLP research, language and cultural documentation, and the preservation of low-resource languages and cultures in AI.
- Organised the [CLRLC-LLMs Workshop @ NeurIPS 2025](#), bringing together researchers, practitioners, and community partners working on low-resource language technologies.
- Leading initiatives focused on multilingual NLP, culturally aware AI systems, and community-centred research collaborations.

CLRLC-LLMs Workshop @ NeurIPS 2025

Workshop Chair

Mexico City, Mexico

September 2025

- Established an international research workshop accepted at NeurIPS 2025 focused on advancing low-resource language technologies and cultural representation in large language models (LLMs).
- Coordinated the program committee, invited speakers, and interdisciplinary collaborations across academia and industry.
- Designed the technical workshop structure, including speaker sessions, lightning talks, and community engagement activities.

Deep Learning Indaba Ideathon 2025

Reviewer

Remote

August 2025

- Reviewed AI/ML project submissions and proposals, ensuring fairness, research quality, and technical relevance.
- Evaluated innovative AI systems across research and applied domains within the African AI ecosystem.
- Certification of Contribution: [View Certificate](#)

Tonative

Training Manager

Lagos, Nigeria

January 2026 – Present

- Designed and implemented training curriculum for participants across diverse linguistic and cultural contexts.
- Reviewed and validated tutors' teaching materials to ensure curriculum alignment and program quality.
- Coordinated learner feedback and performance assessments to improve training effectiveness and engagement.
- Conducted training sessions and mentorship in AI, NLP, and low-resource language technologies. [View Teaching](#)

LevelUp Techies Bootcamp

AI Mentor

Remote

September 2024 – November 2024

- Guided 50+ participants in Python programming and machine learning fundamentals through mentorship and technical support.
- Supported learners in developing foundational AI and data science skills for real-world applications.

PROFESSIONAL DEVELOPMENT

Advanced Language Processing School (ALPS 2026) – Accepted Participant Centre Paul Langevin, French Alps
Competitive selection for an advanced research school in language processing March 29 – April 3, 2026

PROJECTS

Evaluating LLM Performance on Nigerian Pidgin Sentiment Remote
Researcher / NLP Engineer 2026

- Benchmarked GPT-4o-mini on the NaijaSenti corpus for Nigerian Pidgin sentiment classification.
- Designed zero-shot prompts informed by Pidgin linguistics knowledge.
- Built evaluation pipelines with visualisations and performance analysis.

Yorùbá Proverb Classification Remote
Researcher / NLP Engineer July 2024 – September 2024

- Investigated semantic classification of culturally grounded Yorùbá proverbs.
- Compared Naive Bayes, BERT, and AfroLM models for proverb interpretation.
- Conducted morpho-semantic analysis and structured error evaluation.

Language Documentation: Syllable Structure and Phonotactics of Arogbo-Ijaw, Nigeria Ondo, Nigeria
Research Assistant, Obafemi Awolowo University June 2024

- Conducted fieldwork to describe Arogbo-Ijaw syllable patterns using ELAN and Praat.
- Developed annotated language resources for preservation and NLP applications.

English–Portuguese NMT Remote
Researcher / Developer June 2025

- Fine-tuned Helsinki MarianMT transformer for English–Portuguese machine translation.
- Evaluated translation quality using BLEU benchmarking metrics.

Lexicography Project (EN–Yorùbá) Obafemi Awolowo University
Compiler / Research Assistant January 2024 – April 2025

- Compiled a bilingual ecology and climate dictionary for English and Yorùbá.
- Conducted lexical documentation and terminology research.

Automatic Essay Scoring (AES) Remote
Machine Learning Engineer June 2025

- Fine-tuned DistilBERT for automated essay scoring tasks.
- Evaluated model performance using RMSE and Pearson correlation.

Speech Emotion Recognition Remote
Deep Learning Engineer January 2024 – March 2024

- Built an LSTM-based speech emotion recognition system using MFCC and chroma features.
- Deployed an interactive Streamlit application for inference and testing.

TEACHING EXPERIENCE

Data Science Nigeria Remote
Deep Learning / AI Co-Lead January 2025 – November 2025

- Tutored and guided team members in deep learning concepts and model development, delivered hands-on sessions on neural networks, CNNs, and model evaluation that empowered 20+ participants to deploy deep learning models.
- Contributed to planning and executing national deep learning events and hackathons.
- Supported bootcamp management and mentorship programs for emerging AI talents.

Women in Data Science (WiDS) Remote
AI / Machine Learning Lead October 2024 – November 2025

- Led AI/ML workshops for women and coordinated technical learning sessions.
- Designed structured learning paths and created educational resources for all skill levels.
- Organized and moderated speaker sessions promoting inclusivity and mentorship in AI.

LevelUp Techies Bootcamp Remote
AI Mentor September 2024 – November 2024

- Taught Python programming and machine learning fundamentals, upskilling over 50 participants.

AWARDS

NeurIPS 2025 Scholar Award

Neural Information Processing Systems (NeurIPS)

Mexico City, Mexico
2025

- Received the NeurIPS Scholar Award, covering accommodation and full conference registration.
- Granted five complimentary NeurIPS registration tickets as Workshop Chair to support organisers and invited speakers.

WiML @ NeurIPS 2025 Travel Funding Award

Women in Machine Learning (WiML) Workshop at NeurIPS 2025

Mexico City, Mexico

Deep Learning Indaba 2025 Travel Grant

Deep Learning Indaba Conference

Kigali, Rwanda
2025